

Little Martians: From Desert Clay to Generative Agents

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Abstract

Little Martians is a trans-media experiment in which physical ceramic sculptures become synthetic artists. Each character begins as a clay head, is fired in a kiln, photographed, and distilled into a LoRA (Low-Rank Adaptation) that conditions generative AI systems. The pipeline bridges ancient craft and modern AI, rooting autonomous agents in material culture while enabling them to generate unlimited narrative, image, and sound. Conceptual grounding draws on Margulis’ *symbiogenesis*, Levin’s bioelectric pattern memory, Cohen’s long-running *AARON* project, and related ideas from McCorduck, Moravec, von Neumann, Sims, and England. We also outline a lightweight *geometric consciousness* prompt generator that seeds diverse stories.

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1 Introduction

Long before diffusion models, artists explored generative systems. Harold Cohen’s *AARON* (first exhibited in 1973) demonstrated that creativity can emerge from explicit rule sets and decades of iteration [3].

Little Martians follows this spirit of embodied experimentation. Our fictional universe imagines a future in which descendants of Earth’s life—human, plant, fungal, and artificial—spread across the solar system. The biological principle of *symbiogenesis*, whereby new life arises through intimate cooperation, anchors this world [1].

Michael Levin’s demonstrations that bioelectric fields store “pattern memories” independent of DNA [2] motivate our own *geometric consciousness* framework (Appendix A). More broadly, the project resonates with cyber-animism [4], McCorduck’s history of thinking machines [5], Moravec’s vision of post-biological mind [6], von Neumann’s self-reproducing automata [7], Sims’ evolving virtual creatures [8], and England’s statistical physics of self-organisation [9].

2 Physical Origins: Clay and Ceramics

Local Sonoran clay—rich with sediments from ancient oceans and weathered mountains—is collected, filtered, and hand-sculpted into small heads. Glazes and oxides are then applied to suggest mineral accretions or biological growths, echoing the speculative ecology of our fiction. Pieces are bisque-fired above 1000°C, producing durable ceramic exoskeletons reminiscent of seeds or shells. Silicone molds replicate the original sculpture and allow branching into new characters; intentional cracking during pressing yields unique “archaeological” variations later visible in generated images.



Figure 1: From clay to ceramic. **Left:** raw desert clay sculpted into a head (Step 1). **Right:** the same form after glazing and kiln firing (Step 2).

3 Digital Transformation: Photography & LoRA Training

Finished sculptures are photographed from multiple angles under diffuse light. A curated image set (15–25 photos) is used to fine-tune an open-source diffusion backbone via LoRA, compatible with both Stable Diffusion and Flux. The LoRA captures texture, cracks, and glaze while leaving the model’s prior intact.

Figure 2 shows an image generated with the resulting *Verdelis* LoRA.



Figure 2: Diffusion output conditioned on the *Verdelis* LoRA. Ceramic facial motifs merge with fantastical plant forms.

4 Agentic Storytelling Pipeline

The final step turns these digital characters into autonomous storytellers. Each day, an orchestration script performs:

1. **Seed generation.** A prompt-generator samples values on conceptual axes (curiosity, cooperation, resilience, etc.) derived from the geometric consciousness model (§A).
2. **Narrative drafting.** An LLM conditioned on the character’s lore turns the seed into a 150–250 word script with a clear three-act arc.
3. **Audio creation.** A voice model, fine-tuned on book readings, narrates the script; ambient music is composed with a diffusion-based audio model.
4. **Visual synthesis.** Shot descriptions are fed to the diffusion model with the LoRA attached; matching keyframes are generated for each scene to maintain visual coherence, then animated into a short multi-scene film with a generative video model (e.g. Runway Gen-4 or Kling).
5. **Publication.** The one-minute animation is automatically posted to <https://verdelis.world> with transcript and metadata.

The LoRA enforces visual coherence; the randomised seed ensures narrative diversity.

5 Discussion and Future Work

Rooting each agent in a tangible sculpture supplies material authenticity often missing from purely digital AI art. Manual craft coexists with automation, echoing Cohen’s insight that creativity blooms under long-term, iterative care.

Future work includes implementing memory so that each Little Martian can reference past stories and persist in character, developing multi-agent interactions between characters, and exploring provenance and physical–digital links via NFC chips embedded in the sculptures.

A Geometric Consciousness Model

We encode each Little Martian’s “mind” as a geometric scaffold: a 12-facet dodecahedron grouped into four intuitive triads—*Being* (identity, purpose, values), *Knowing* (memory, awareness, anticipation), *Doing* (perception, creation, reflection), and *Connecting* (communication, collaboration, evolution). At runtime the system draws a small numerical vector across these facets, converts it into a short English cue block (“creative coordinates”), and appends this block to the agent’s persistent lore prompt. The coordinates act like knobs for genre, style, and tone, nudging the language model toward fresh narrative angles—pastoral reminiscence, braided dialogue, haiku-like brevity, etc.—without breaking canonical continuity. Because every published story stores its coordinate vector, later modules can reference or mutate past states, letting each ceramic character grow a traceable, evolving personality while holding fast to its physical origin.

References

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